

App Development Project Report

App Dev Project Report

1. Student Details

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About Me: I am a student who is enrolled in IIT Madras Online BS in Data Science and Applications. I have deep interest in machine learning, deep learning and Artificial intelligence.

2. Project Details

Project Title: Arohan - Trekking Manage App

Problem Statement:

To design and build a web-based application that allows **Admin, Trek Staff, and Users (Trekkers)** to interact with the system based on their roles.

Approach:

The application was developed using Flask as the core backend framework, following a template-driven architecture. It employs a custom role-based access control system to serve three distinct user types: Admins, Staff, and Trekkers. The backend manages database operations via SQLAlchemy, ensuring real-time updates to trek capacities, while the frontend dynamically renders this data using Jinja2 templates. .

3. AI/LLM Declaration

I used **Gemini** to assist in Flask routing errors and errors related to SQLALCHEMY database lock and some 4-5 backend functions suggestions(total 26 functions are there) in [app.py](#)(act as controller for my web application) . The extent of of AI LLM usage is around 17-18%.limited mostly to troubleshooting error, code suggestion,

Front-end development is 95% done by me, less than 5% AI was used there only for templating where the function of [app.py](#) look somewhat different than other functions. As it was mostly templating related.

4. Technologies and Frameworks Used

Technology Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja	Template engine for rendering dynamic HTML pages
Bootstrap	Very Limited use, only used in main.html file(act as index page of my web application where login and register button was their only with heading of app)
SQLite	Lightweight local database for storing user data

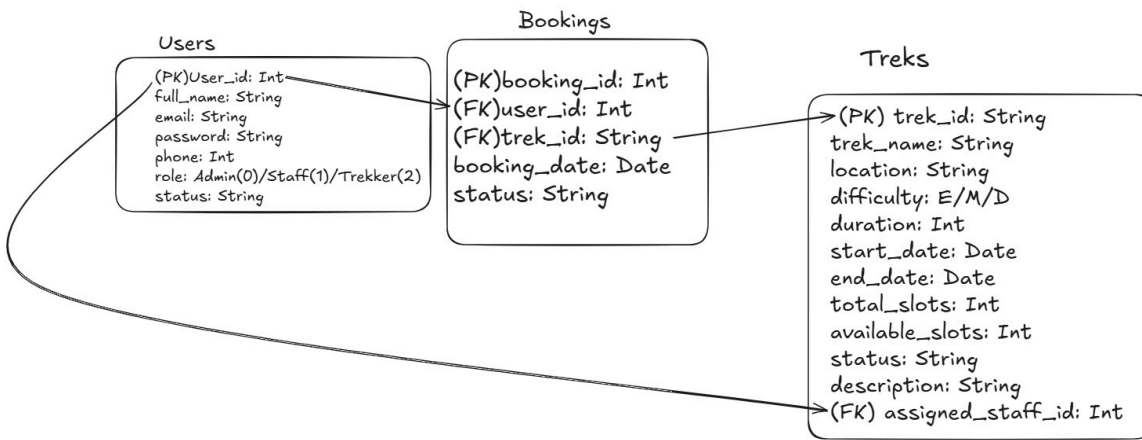
5. Database Schema / ER Diagram

Tables:

1. **User** — stores user profile details, roles and account status
2. **Treks**— stores trek_information, dates and allotment size
3. **Booking**— log trek reservations and links trekker to specific treks

Relationships:

- One-to-Many → `User(Trekkers) -> Bookings(One user can make multiple bookings)`
- One-to-Many → `Treks -> Bookings(One trek can have multiple booking from different users)`



I made it using Excalidraw website which was recommended in live classes by IIT madras bs program teachers in mad 1 project course

6. API Resource Endpoints

No API was there in my Web application as it was optional in App Dev I Project Statement.

7. Architecture and Features (optional)

Architecture Overview:

My web application uses Model View Controller(MVC) architectural pattern.

- **app.py(Controller)** – act as controller for our web application, it handles all HTTP routing, user session management, and business logic.
- **/models(Model)** – contains database schema and relationships. The tables were build using SQLAlchemy to manage data persistence.
- **/templates(View)** – Jinja2 HTML templates which are dynamically rendered by the server based on data based from controller

- `/static(Assets)` – CSS

Implemented Features:

- **User Registration and Login:** Secure authentication system with manual user session handling via Flask.
- **Role based control:** Distinct routing and permissions for Admins (0), Staff (1), and Trekkers (2).
- **Dynamic Trek Booking:** Trekkers can book available treks, which automatically updates and decrements the available slots in the database.
- **Trek Management Dashboard:** Admins and assigned Staff can manage trek details, including start/end dates, location, and difficulty.
- **Dynamic UI Rendering:** Server-side rendered HTML interfaces using Jinja2, utilizing custom CSS and limited Bootstrap for layout.
- **Real-time Validation:** Prevents users from booking treks that are full or inactive, displaying clear error messages.

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1l8Z_3AZBkfvUI4lYTv9UgOqMI0_8d-gZ/view?usp=sharing
